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Analysis of I–V Characteristics and Leakage in CMOS

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1. Experiment Overview

The objective of this experiment is to investigate the current–voltage (I–V) characteristics and leakage behavior of a CMOS transistor using LTspice simulation. The I–V characteristics describe the relationship between the applied voltage and the resulting current flowing through a MOSFET, providing valuable information about the operating regions of the device. In modern integrated circuits, leakage current has become an increasingly important parameter due to continuous transistor scaling and the demand for low-power electronic systems. Through simulation, the electrical behavior of CMOS devices can be analyzed without the need for physical fabrication, allowing designers to evaluate device performance accurately during the design stage.

2. Objective

The objectives of this experiment are:

- To simulate the I–V characteristics of CMOS transistors using LTspice.
- To analyze the relationship between gate voltage, drain voltage, and drain current.
- To observe the leakage current under different biasing conditions.
- To understand the effects of transistor operation in the cutoff, linear, and saturation regions.
- To evaluate the importance of leakage current in low-power CMOS circuit design.

3. Theory

Complementary Metal-Oxide-Semiconductor (CMOS) technology is the dominant technology used in modern integrated circuits because of its high noise immunity, low static power consumption, and excellent scalability. A CMOS device consists of complementary NMOS and PMOS transistors, where only one transistor conducts during steady-state operation, resulting in extremely low static power dissipation.

The electrical characteristics of a MOSFET are commonly represented by its I–V curves, which illustrate how the drain current (ID) varies with changes in gate-to-source voltage (VGS) and drain-to-source voltage (VDS).

These characteristics determine the operating regions of the transistor:

- **Cutoff Region:** The transistor remains OFF because the gate voltage is below the threshold voltage, producing only a very small leakage current.
- **Linear (Triode) Region:** The transistor behaves like a voltage-controlled resistor, and the drain current increases approximately linearly with the drain voltage.
- **Saturation Region:** The transistor operates as a current source where the drain current becomes relatively independent of drain voltage and is mainly controlled by the gate voltage.

Leakage current refers to the small amount of current that flows even when a transistor is intended to be OFF. As CMOS technology scales into deep submicron dimensions, leakage mechanisms such as subthreshold leakage, gate oxide tunneling, and reverse-biased junction leakage become increasingly significant. Although these currents are very small compared to the ON-state current, they contribute to static power consumption and reduce battery life in portable electronic devices. Therefore, minimizing leakage current is an important consideration in modern VLSI design.

4. Software Used

- LTspice 26.0.1
- 65 nm CMOS Technology Model
- Windows Operating System

5. Methodology

The 65 nm CMOS transistor model was successfully implemented and biased within the LTspice environment. To establish the required operating conditions, appropriate DC voltage sources were connected directly to the gate, drain, source, and substrate terminals. A primary DC sweep analysis was then performed by systematically varying the drain voltage across a range of fixed gate voltage levels. The resulting drain current was recorded to generate the transistor's definitive I–V characteristic curves. A second, distinct simulation sequence was carried out to evaluate the device while it remained strictly in the OFF state. This was achieved by applying a gate voltage safely below the established threshold voltage of the 65 nm node. LTspice automatically plotted these simulation results, providing a clear visual representation of the voltage-current relationships. The finalized data allowed for a thorough analysis of the transistor's behavior

across its distinct operating regions. Ultimately, special attention was dedicated to quantifying the exact magnitude of the leakage current under these low-voltage conditions.

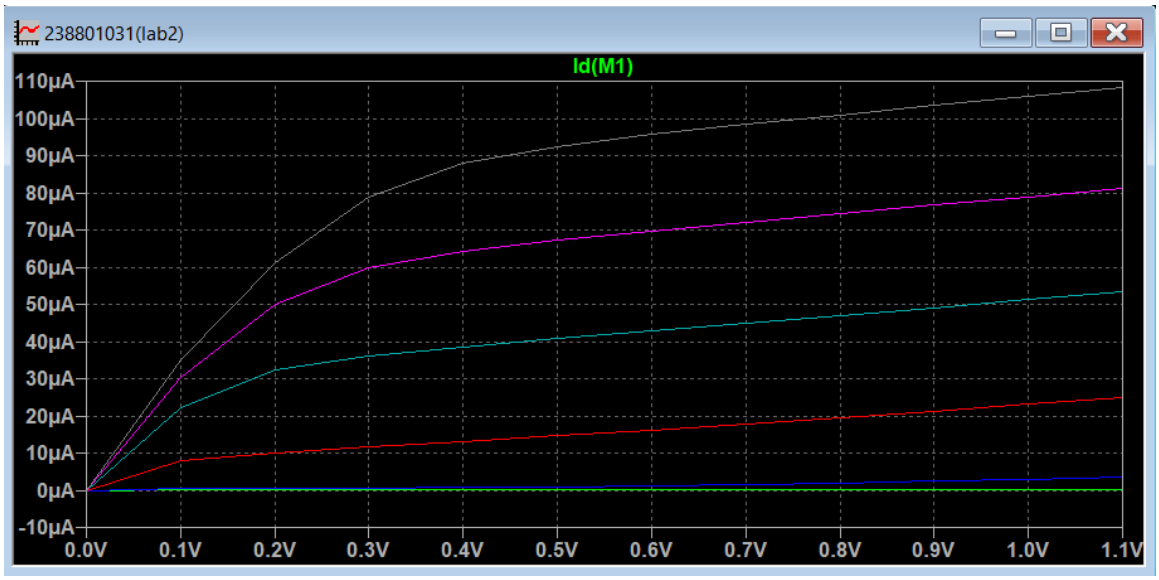
6. Programs

I–V Characteristics of 65nm NMOS:

238801031(lab2)

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab2).sp
*I-V characteristics of 65nm NMOS
M1 2 1 0 0 nmos w=100n l=65n m=1
VDS 2 0 1
VGS 1 0 0
.DC VDS 0 1.1 0.1 VGS 0.1 1.1 0.2
*model
.lib'65nm.pm'
.end
```

I–V Curves:



Leakage Effect Analysis:

238801031(lab2)

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab3).sp
*Part 1 - Study of all leakage effects
*65nm min-size NMOS

M1 2 1 0 0 nmos W=100n L=65n

VDS 2 0 {Vds}
VGS 1 0 0

.dc VGS -0.5 1.1 0.1
.step param Vds list 0.1 0.5 1.1

.inc "65nm.pm"
.end
```

Leakage Effect Curves:



Temperature Effect on Nmos:

238801031(lab3)

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab3).sp
*Temperature Effects on NMOS
M1 2 1 0 0 nmos W=100n L=65n
VDS 2 0 1.1
VGS 1 0 0
.dc VGS -0.5 1.1 0.01
.step TEMP list -40 30 100
.lib 65nm.pm
.end
```

Temperature Effect Curve:



Body Biasing Effect:

238801031(lab3)

```
* C:\Users\Sifat\Desktop\ltspice\238801031(lab3).sp
*Body Biasing Effect on NMOS
M1 2 1 0 B nmos W=100n L=65n

VDS 2 0 1.1
VGS 1 0 0
VBS B 0 {Vbody}

.dc VGS -0.5 1.1 0.01
.step param Vbody list -0.5 0 0.5
.lib 65nm.pm
.end
```

Body Biasing Effect Curve:



7. Result:

The LTspice simulation successfully generated the I–V characteristics and leakage current behavior of the CMOS transistor. The simulation verified the expected operating regions of the MOSFET and demonstrated the relationship between gate voltage, drain voltage, and drain current. Leakage current was observed in the OFF state, confirming that practical CMOS devices exhibit small but measurable static current due to intrinsic device physics.

8. Conclusion:

This experiment successfully demonstrated the analysis of CMOS transistor I–V characteristics and leakage current using LTspice simulation. The results confirmed the theoretical operating behavior of CMOS transistors in the cutoff, linear, and saturation regions while illustrating the existence of leakage current under OFF-state conditions. The experiment also emphasized the importance of leakage analysis in modern semiconductor technology, where reducing static power consumption is essential for improving circuit efficiency, reliability, and battery life. LTspice proved to be an effective simulation platform for understanding MOSFET characteristics and supporting the design and optimization of CMOS integrated circuits.